

Natural immunomodulators and their stimulation of immune reaction: true or false?



Natural immunomodulators are getting more and more popular. The popularity, however, often brings over-optimistic claims and mediocre effects. The purpose of the present study was to directly compare eleven most commonly used immunomodulators. Through testing both cellular and humoral branches of immune reactions, we found that most of the immunomodulators tested have limited, if any, effects, with glucan being consistently the most active molecule strongly stimulating every reaction evaluated. These data were also confirmed using a Lewis lung cancer model, where only glucan and resveratrol lowered the number of metastases.